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BASIC MATHEMATICS

FORM TWO

Form Two National Assessment (FTNA) 2024 Answer Key and Worked Solutions

Subject	Basic Mathematics
Code	041
Level	Form Two
Year	2024
Exam Board	NECTA
Exam	Form Two National Assessment (FTNA)
Type	Answer key and worked solutions
Questions	10 compulsory questions; 21 answered parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: BasicMath-F2-2024.pdf

The scanned pages containing questions 4 and 5, and later parts of 7 and 8, appear out of page order. Solutions follow the printed question numbers.

Question 1 — Factors and estimation

1(a). Find the GCF and LCM of 90 and 240.

Step 1: Prime factors: $90 = 2 \times 3^2 \times 5$ and $240 = 2^4 \times 3 \times 5$.

Step 2: GCF uses minimum shared powers: $2 \times 3 \times 5 = 30$. LCM uses maximum powers: $2^4 \times 3^2 \times 5 = 720$.

Final Answer: GCF = 30; LCM = 720.

1(b). Estimate $8108 \div 37$.

Step 1: Round 8108 to 8000 and 37 to 40 for an easy estimate.

Step 2: $8000 \div 40 = 200$.

Final Answer: Approximately 200, using one-significant-figure estimation.

Study Tip: No precision is specified. A compatible-number estimate $8140 \div 37 = 220$ is also reasonable and closer to the exact quotient, about 219.135. State the approximation you use.

Section Summary: Follow the operation and notation shown in the original paper. Keep fractions or radicals exact, and round only when an accuracy is requested.

Question 2 — Fractions and recurring decimals

2(a). Simplify $3 \frac{9}{10} \div (3 \frac{3}{5} - 1 \frac{1}{2})$.

Step 1: Convert mixed numbers: $3 \frac{9}{10} \div (1 \frac{18}{5} - 1 \frac{1}{2})$.

Step 2: Bracket first: $1 \frac{18}{5} - 1 \frac{1}{2} = \frac{36}{10} - \frac{15}{10} = \frac{21}{10}$.

Step 3: Divide by multiplying the reciprocal: $(\frac{39}{10}) \times (\frac{10}{21}) = \frac{39}{21} = \frac{13}{7}$.

Final Answer: $\frac{13}{7} = 1 \frac{6}{7}$.

2(b). Change 1.23 to a mixed number; the block 23 repeats.

Step 1: Let $x = 1.232323\ldots$. Then $100x = 123.232323\ldots$.

Step 2: Subtract x from $100x$: $99x = 122$, so $x = 122/99 = 1 \frac{23}{99}$.

Final Answer: 1 23/99.

Section Summary: Follow the operation and notation shown in the original paper. Keep fractions or radicals exact, and round only when an accuracy is requested.

Question 3 — Interest and units

3(a). How much must be lent to earn Tsh 36,000 simple interest at 5% per annum in 6 months?

Step 1: Use $t = 6/12 = 1/2$ year and $I = Prt/100$.

Step 2: $P = 100I/(rt) = (100 \times 36,000)/(5 \times 1/2) = 1,440,000$.

Final Answer: Tsh 1,440,000.

3(b). Nyanjara bought 50 bottles of milk, each holding 300 ml, for 70 children. Find the total litres bought.

Step 1: Total volume = $50 \times 300 = 15,000$ ml.

Step 2: Since $1000 \text{ ml} = 1 \text{ litre}$, $15,000 \text{ ml} = 15 \text{ litres}$; the number of children does not change the volume bought.

Final Answer: 15 litres.

Section Summary: Follow the operation and notation shown in the original paper. Keep fractions or radicals exact, and round only when an accuracy is requested.

Question 4 — Angles and perimeter

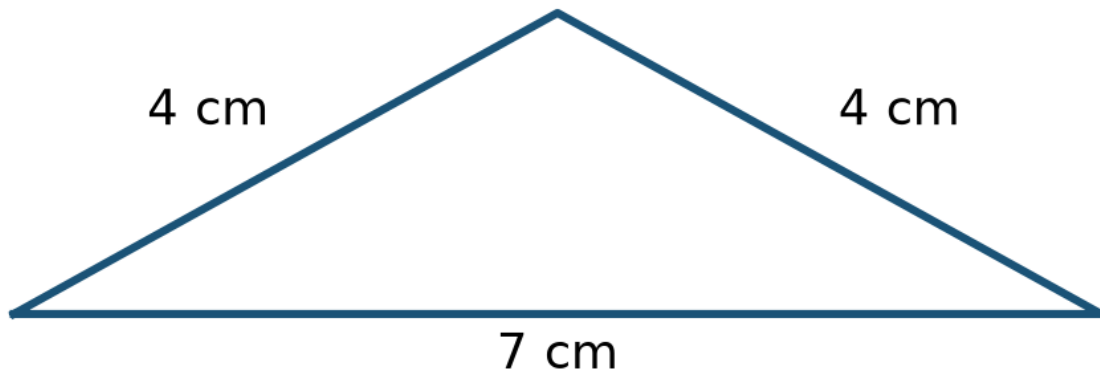
4(a). One of two complementary angles is twice the other. Find both.

Step 1: Let the smaller angle be t and the larger $2t$. Complementary angles sum to 90° .

Step 2: $t + 2t = 90$ gives $t = 30$; the other is 60 .

Final Answer: 30° and 60° .

4(b). An isosceles triangle has perimeter 15 cm and base 7 cm. Draw it and find each equal side.



Step 1: Let each equal side be s . Then $2s + 7 = 15$.

Step 2: $2s = 8$, so $s = 4$ cm. The side lengths satisfy $4 + 4 > 7$, so the triangle is possible.

Final Answer: Each equal side is 4 cm.

Section Summary: Follow the operation and notation shown in the original paper. Keep fractions or radicals exact, and round only when an accuracy is requested.

Question 5 — Algebra

5(a). Solve $1 - (x + 2)/2 = x - 3$.

Step 1: Multiply every term by 2: $2 - (x + 2) = 2x - 6$.

Step 2: Simplify: $-x = 2x - 6$, so $3x = 6$ and $x = 2$. Check both sides equal -1 .

Final Answer: $x = 2$.

5(b). What term must be added to $n^2 + (1 \frac{1}{2})n = 0$ to complete the square?

Step 1: The coefficient of n is $3/2$. Half of it is $3/4$.

Step 2: Square this half: $(3/4)^2 = 9/16$. Then $n^2 + (3/2)n + 9/16 = (n + 3/4)^2$.

Final Answer: Add $9/16$ to both sides; $(n + 3/4)^2 = 9/16$.

Section Summary: Follow the operation and notation shown in the original paper. Keep fractions or radicals exact, and round only when an accuracy is requested.

Question 6 — Translation and coordinates

6(a). A translation maps $P(-2,2)$ to $P'(0,0)$. Find the image of $(5,-1)$.

Step 1: Translation vector = $(0 - (-2), 0 - 2) = (2, -2)$.

Step 2: Add the vector to $(5, -1)$: $(5 + 2, -1 - 2) = (7, -3)$.

Final Answer: $(7, -3)$.

6(b). A line passes through $(1,2)$ and cuts the y -axis at $(0,2)$. Find its equation.

Step 1: Gradient = $(2 - 2)/(1 - 0) = 0$.

Step 2: A zero-gradient line is horizontal; its constant y -coordinate is 2.

Final Answer: $y = 2$.

Section Summary: Follow the operation and notation shown in the original paper. Keep fractions or radicals exact, and round only when an accuracy is requested.

Question 7 — Radicals and logarithms

7(a). If $x = \sqrt{3}$, $y = \sqrt{2}$ and $z(x - y) = 2$, write $z = a(\sqrt{b} + \sqrt{c})$.

Step 1: Substitute: $z = 2/(\sqrt{3} - \sqrt{2})$.

Step 2: Rationalise with $\sqrt{3} + \sqrt{2}$: $z = 2(\sqrt{3} + \sqrt{2})/(3 - 2) = 2(\sqrt{3} + \sqrt{2})$.

Final Answer: $z = 2(\sqrt{3} + \sqrt{2})$; $a = 2$, $b = 3$, $c = 2$.

7(b). Simplify $(\log 8 - 2 \log 4)/(\log 4 - \log 2)$.

Step 1: Use $\log 8 = 3 \log 2$ and $\log 4 = 2 \log 2$.

Step 2: Numerator = $3 \log 2 - 4 \log 2 = -\log 2$; denominator = $2 \log 2 - \log 2 = \log 2$.

Step 3: Divide: $-\log 2/\log 2 = -1$.

Final Answer: -1 .

Section Summary: Follow the operation and notation shown in the original paper. Keep fractions or radicals exact, and round only when an accuracy is requested.

Question 8 — Similarity and congruence

8(a). $\triangle PQR \sim \triangle LMN$, $\angle MNL = 40^\circ$ and $\angle QPR = 60^\circ$. Find $\angle RQP$.

Step 1: The ordered correspondence is $P \leftrightarrow L$, $Q \leftrightarrow M$ and $R \leftrightarrow N$; therefore $\angle PRQ = 40^\circ$.

Step 2: In $\triangle PQR$, $\angle RQP = 180 - 60 - 40 = 80^\circ$.

Final Answer: 80° .

8(b). In the figure E,D,C,B are collinear, $AE = AB$ and $ED = CB$. Prove $\triangle ABC \cong \triangle AED$.

Step 1: $AE = AB$ makes the large triangle AEB isosceles, so its base angles $\angle AEB = \angle EBA$.

Step 2: Because D and C lie on EB, $\angle AED = \angle ABC$. Also $ED = BC$ and $AE = AB$ are given.

Step 3: Two sides and the included angle match, proving SAS congruence with $A \leftrightarrow A$, $B \leftrightarrow E$, $C \leftrightarrow D$.

Final Answer: $\triangle ABC \cong \triangle AED$ by SAS.

Section Summary: Follow the operation and notation shown in the original paper. Keep fractions or radicals exact, and round only when an accuracy is requested.

Question 9 — Pythagoras and exact trigonometry

9(a). A student travels 14 km east, then south to school. Home-school direct distance is 20 km. Find junction-school distance to 3 decimal places.

Step 1: The east and south legs are perpendicular. Let the south leg be d .

Step 2: Pythagoras: $d^2 = 20^2 - 14^2 = 400 - 196 = 204$.

Step 3: $d = \sqrt{204} \approx 14.282856\dots$, rounding to 14.283 km.

Final Answer: 14.283 km.

9(b). Evaluate $2 \sin 60^\circ + \cos 30^\circ - \tan 60^\circ$ in radical form.

Step 1: Use $\sin 60^\circ = \sqrt{3}/2$, $\cos 30^\circ = \sqrt{3}/2$, $\tan 60^\circ = \sqrt{3}$.

Step 2: $2(\sqrt{3}/2) + \sqrt{3}/2 - \sqrt{3} = \sqrt{3}/2$.

Final Answer: $\sqrt{3}/2$.

Section Summary: Follow the operation and notation shown in the original paper. Keep fractions or radicals exact, and round only when an accuracy is requested.

Question 10 — Sets and pie charts

10(a). Venn regions are A only = $7 - 3x$, both = x , B only = $5 - x$. If $n(A) = n(B)$, find $n(A \cup B)$ and $n(A)$.

Step 1: $n(A) = 7 - 3x + x = 7 - 2x$; $n(B) = 5 - x + x = 5$.

Step 2: Equate: $7 - 2x = 5$, so $x = 1$. Regions are then 4,1,4.

Step 3: Union = $4 + 1 + 4 = 9$; A total = $4 + 1 = 5$.

Final Answer: $x = 1$; $n(A \cup B) = 9$; $n(A) = 5$.

10(b)(i). John's Monday pie chart shows sleeping 105° , school 90° , meals 30° , games 60° and others 45° . How many hours for private study?

Step 1: Known angles total $105 + 90 + 30 + 60 + 45 = 330^\circ$. Private study = $360 - 330 = 30^\circ$.

Step 2: A day has 24 hours; private-study time = $(30/360) \times 24 = 2$ hours.

Final Answer: 2 hours.

10(b)(ii). How many hours does John sleep?

Step 1: Sleeping accounts for 105° of the whole 360° day.

Step 2: Sleeping time = $(105/360) \times 24 = 7$ hours.

Final Answer: 7 hours.

Section Summary: Follow the operation and notation shown in the original paper. Keep fractions or radicals exact, and round only when an accuracy is requested.